

ADITYA OJHA

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SUMMARY

Software Engineer with **hands-on experience** in **backend development**, **systems programming**. Strong foundation in **data structures and algorithms**, **operating systems**, and **multithreaded programming**, with hands-on experience in modern cloud and AI workflows.

EXPERIENCE

Oracle

Associate Software Developer Intern

Hyderabad, Telangana

January 2026 – Present

- Mastered the legacy **C++ Session Border Controller (SBC)** architecture along with its **Docker** and **Kubernetes** infrastructure stack.
- Resolved **500+ compiler warnings** and refactored **400+ lines** of production code, improving maintainability and accelerating compilation through successful Change Requests.
- Used **GDB** for **multithreaded debugging** to identify and patch critical logical vulnerabilities, preventing runtime failures and ensuring **memory safety**.

ITJobxs.com

Software Development Engineer Intern

Remote

January 2025 – February 2025

- Implemented **PHP validation** with **Google reCAPTCHA**, reducing bot registrations by **90%** and saving **5 engineering hours/week**.
- Developed a **MERN-stack** e-commerce platform with **JWT** authentication, improving verification speed by **3X**.

PROJECTS

Automated Multi-Agent Job Pipeline | Python, LangGraph, Kafka, PostgreSQL, Docker

May 2026

- Architected a **fault-tolerant, event-driven browser ecosystem** using **LangGraph**, **Playwright**, and **Kafka**, reducing manual workflow overhead by **85%**.
- Built a deterministic keyword extraction microservice using the **Aho-Corasick algorithm**, reducing **LLM token extraction costs** by **60%** through optimized contextual injection.
- Developed a **Human-in-the-Loop (HITL) WebSocket protocol** and automated **LaTeX resume generation pipeline** to dynamically create tailored resume variants in real time.

Encrypty | C++, POSIX Threads, Shared Memory, Semaphores

March 2025

- Designed a **process-thread hybrid encryption pipeline** achieving a **5X speedup** over single-threaded baselines using contention-aware queueing and synchronization primitives.
- Implemented parallel file chunk processing using **POSIX Threads**, **shared memory (mmap)**, and **semaphores** to optimize CPU utilization and minimize synchronization overhead.

CF Stalker | React, Tailwind CSS, Express.js, MongoDB

April 2025

- Built a **Codeforces analytics platform** with real-time backend pipelines, scaling to **1,000+ users**.
- Integrated **Codeforces APIs** to visualize user rating trends, problem-solving patterns, topic distributions, and contest performance through interactive dashboards.

TECHNICAL SKILLS

Languages: C/C++ (20/23), Java, Python, JavaScript/TypeScript, SQL, Bash/Shell

Backend & Web: Node.js, Express, FastAPI, Spring Boot, React.js, RESTful APIs

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, Git, Jenkins, CI/CD, Linux/UNIX

Databases & Messaging: Oracle DB, MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka

Architecture: System Design, Microservices, Multithreading, Asynchronous I/O, Object-Oriented Design, Data Structures and Algorithms, Operating Systems, Computer Networks

AI & Testing: Generative AI, LangGraph, Playwright, Retrieval-Augmented Generation (RAG), JUnit, End-to-End (E2E) Testing

ACHIEVEMENTS

Competitive Programming: Leetcode Guardian (Peak 2230, Top 0.9%), Codeforces Expert (Peak 1710), CodeChef 5-Star (Peak 2023)

Global Competitions: ICPC Regional Finalist (AIR 71 Prelims, AIR 149 Amritapuri), Amazon Hackon 2025 (Top 130 of 18,000+ teams)

EDUCATION

National Institute of Technology, Raipur

B.Tech in Computer Science and Engineering; CGPA: 8.00/10

Raipur, Chhattisgarh

2022 - 2026